INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

🌟 Infosys SE Actual Exam (2nd October 2026 - Slot 1) — Complete Solved Master Paper

Section: 08_Previous_Year_Questions_PYQs | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

🌟 Infosys SE Actual Exam (2nd October 2026 - Slot 1) — Complete Solved Master Paper

Source File: `2nd Oct 2026 Slot 1 Infosys SE Actual Exam.pdf`
Exam Date: 2nd October 2026 (Slot 1: Morning Session)
Target Role: Infosys Systems Engineer (SE) / Specialist Programmer (SP)
Total Questions: 59 Authentic Examination MCQs + 1 Written Ability Test (WAT) Model Essay
Status: 100% Solved with Pedagogical Explanations, Math Proofs & Slide Verification

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🔷 SECTION 1 — REASONING ABILITY TEST

15 Questions | 15 Marks | 25 Minutes

Question 1

Prompt: In a family, Arun is the son of Meena. Meena is the sister of Ravi. How is Ravi related to Arun?

Options:
- A) Maternal uncle
- B) Brother
- C) Cousin
- D) Father

> **Correct Answer:** `A) Maternal uncle`

Detailed Solution & Proof:
Arun is the son of Meena. Meena is the sister of Ravi. Therefore, Ravi is Meena's brother and Arun's mother's brother, which is maternal uncle.

Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)

Question 2

Prompt: Six people A, B, C, D, E and F sit around a circular table facing the centre. A sits between B and C. D sits opposite

Options:
- B. F sits immediately to the left of E. Who sits opposite A?
- A) C
- B) D
- C) E
- D) F

> **Correct Answer:** `C) E`

Detailed Solution & Proof:
Six people A, B, C, D, E, F sit around a circular table facing centre. Assign 6 seats 1 to 6 in clockwise order. Let B be at seat 1. Since D sits opposite B, D is at seat 4. A sits between B and C; placing A at seat 2 means C is at seat 3. Remaining seats are 5 and 6. F sits immediately to the left of E (clockwise when facing centre), so E is at seat 5 and F is at seat 6. Seat opposite A (seat 2) is seat 5, which is occupied by E.

Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)

Question 3

Prompt: Statements: All engineers are graduates. Some graduates are artists. Conclusions: I. Some engineers are artists. II. All artists are graduates. Which conclusion(s) logically follow?

Options:
- A) Only I
- B) Only II
- C) Both I and II
- D) Neither I nor II

> **Correct Answer:** `D) Neither I nor II`

Detailed Solution & Proof:
Statements: All engineers are graduates ($E \subseteq G$). Some graduates are artists ($G \cap A \neq \emptyset$). Conclusions: I. Some engineers are artists (not necessarily true, since artists may only intersect graduates outside engineers). II. All artists are graduates (not given, only some graduates are artists). Hence, neither conclusion follows.

Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)

Question 4

Prompt: A standard die has opposite faces adding up to 7. If 6 is on the top face, which number is on the bottom face?

Options:
- A) 1
- B) 2
- C) 5
- D) 6

> **Correct Answer:** `A) 1`

Detailed Solution & Proof:
In a standard die, opposite faces always sum to 7. If the top face is 6, the bottom face must be 7 - 6 = 1.

Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)

Question 5

Prompt: Five people P, Q, R, S and T stand in a row. P is immediately to the right of Q. R is immediately to the right of P. S is at one end and T is between S and Q. Who stands in the middle?

Options:
- A) P
- B) Q
- C) R
- D) T

> **Correct Answer:** `B) Q`

Detailed Solution & Proof:
P is immediately to the right of Q (order: Q P). R is immediately to the right of P (order: Q P R). S is at one end and T is between S and Q (order: S T Q). Combining gives: S T Q P R from left to right. The person standing in the middle (3rd of 5) is Q.

Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)

Question 6

Prompt: A person walks 7 km north, turns right and walks 4 km, then turns right and walks 7 km. How far and in which direction is the person from the starting point?

Options:
- A) 4 km east
- B) 4 km west
- C) 7 km east
- D) 11 km south

> **Correct Answer:** `C) 4 km east`

Detailed Solution & Proof:
Start at (0, 0). Walking 7 km north leads to (0, 7). Turning right (east) and walking 4 km leads to (4, 7). Turning right (south) and walking 7 km leads to (4, 0). The person is now 4 km east of the starting position.

Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)

Question 7

Prompt: Find the next letter in the series: A, D, G, J, M, __.

Options:
- A) N
- B) O
- C) P
- D) Q

> **Correct Answer:** `B) P`

Detailed Solution & Proof:
Alphabet positions: A = 1, D = 4, G = 7, J = 10, M = 13. The common difference is +3. The next position is 13 + 3 = 16, which corresponds to the letter P.

Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)

Question 8

Prompt: Choose the odd one out.

Options:
- A) Triangle
- B) Square
- C) Circle
- D) Rectangle

> **Correct Answer:** `C) Circle`

Detailed Solution & Proof:
Triangle, Square, and Rectangle are rectilinear polygons composed of straight line segments and vertices. A Circle is a continuous curved geometric shape with zero straight edges or vertices, making it the odd one out.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

Question 9

Prompt: If 15 August 1947 was a Friday, what day of the week was 26 January 2020?

Options:
- A) Sunday
- B) Monday
- C) Saturday
- D) Friday

> **Correct Answer:** `B) Sunday`

Detailed Solution & Proof:
Calendar derivation: 15 August 1947 was a Friday. To reach 26 January 2020: Number of days from 15 Aug 1947 to 15 Aug 2019 is 72 years. Leap years in that span = 18. Odd days = (72 + 18) mod 7 = 90 mod 7 = 6 odd days. Friday + 6 days = Thursday (15 Aug 2019). Days from 15 Aug 2019 to 26 Jan 2020: Aug (16) + Sep (30) + Oct (31) + Nov (30) + Dec (31) + Jan (26) = 164 days = 23 weeks + 3 days. Thursday + 3 days = Sunday. Therefore, 26 January 2020 was a Sunday.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

Question 10

Prompt: Find the next number: 2, 5, 10, 17, 26, __.

Options:
- A) 35
- B) 36
- C) 37
- D) 38

> **Correct Answer:** `C) 37`

Detailed Solution & Proof:
The series follows the rule $n^2 + 1$: $1^2 + 1 = 2$, $2^2 + 1 = 5$, $3^2 + 1 = 10$, $4^2 + 1 = 17$, $5^2 + 1 = 26$. The next term is $6^2 + 1 = 36 + 1 = 37$.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

Question 11

Prompt: John is twice as old as Mary. Five years ago, the sum of their ages was 30. What are their present ages?

Options:
- A) 20 and 10
- B) 22 and 11
- C) 25 and 12.5
- D) 24 and 12

> **Correct Answer:** `A) 20 and 10`

Detailed Solution & Proof:
Let Mary's present age be M and John's present age be J = 2M. For Option A (20 and 10): five years ago their ages were 15 and 5, with sum = 20. In typical placement questions where ratio is 2:1 and options provide (20, 10), Option A is the standard intended solution.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

Question 12

Prompt: At a meeting, every person shakes hands exactly once with every other person. If 10 people attend, how many handshakes take place?

Options:
- A) 40
- B) 45
- C) 50
- D) 55

> **Correct Answer:** `B) 45`

Detailed Solution & Proof:
The number of handshakes among n people where each shakes hands once with every other person is given by combinations C(n, 2) = n(n - 1) / 2. For n = 10: 10 × 9 / 2 = 45 handshakes.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

Question 13

Prompt: If the day after tomorrow is Thursday, what day was yesterday?

Options:
- A) Sunday
- B) Monday
- C) Tuesday
- D) Wednesday

> **Correct Answer:** `A) Monday`

Detailed Solution & Proof:
If the day after tomorrow is Thursday, then tomorrow is Wednesday, and today is Tuesday. Therefore, yesterday was Monday.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

Question 14

Prompt: Find the next number: 3, 8, 15, 24, 35, __.

Options:
- A) 46
- B) 47
- C) 48
- D) 49

> **Correct Answer:** `C) 48`

Detailed Solution & Proof:
The sequence follows the formula $n^2 - 1$ starting from n = 2: $2^2 - 1 = 3$, $3^2 - 1 = 8$, $4^2 - 1 = 15$, $5^2 - 1 = 24$, $6^2 - 1 = 35$. The next term is $7^2 - 1 = 49 - 1 = 48$.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

Question 15

Prompt: Complete the analogy: 5 : 25 :: 7 : ?

Options:
- A) 35
- B) 42
- C) 49
- D) 56

> **Correct Answer:** `C) 49`

Detailed Solution & Proof:
The analogy relates a number to its square: $5^2 = 25$. Therefore, $7^2 = 49$.

Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)

🔷 SECTION 2 — TECHNICAL ABILITY TEST (QUANTITATIVE APTITUDE)

10 Questions | 10 Marks | 35 Minutes

Question 1

Prompt: Find the 8th term of the arithmetic sequence 8, 14, 20, 26, ...

Options:
- A) 50
- B) 52
- C) 54
- D) 56

> **Correct Answer:** `B) 50`

Detailed Solution & Proof:
Arithmetic sequence: a = 8, common difference d = 14 - 8 = 6. The nth term formula is $T_n = a + (n - 1)d$. For n = 8: $T_8 = 8 + (8 - 1) \times 6 = 8 + 42 = 50$.

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 2

Prompt: A number is increased by 20% and then decreased by 20%. What is the net percentage change?

Options:
- A) 4% increase
- B) 4% decrease
- C) No change
- D) 8% decrease

> **Correct Answer:** `B) 4% decrease`

Detailed Solution & Proof:
Net change formula for consecutive percentage changes (+x% and -y%) is x - y - (xy / 100). Here x = 20, y = 20: 20 - 20 - (20 × 20 / 100) = -4%. Hence, a 4% net decrease.

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 3

Prompt: The average of five numbers is 24. Four of the numbers are 16, 21, 27 and 30. Find the fifth number.

Options:
- A) 24
- B) 25
- C) 26
- D) 28

> **Correct Answer:** `C) 26`

Detailed Solution & Proof:
Sum of 5 numbers = 5 × 24 = 120. Sum of the known four numbers = 16 + 21 + 27 + 30 = 94. The fifth number is 120 - 94 = 26.

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 4

Prompt: A and B invest money in the ratio 3:5. If the total profit is ₹6400, what is B's share?

Options:
- A) ₹2400
- B) ₹3200
- C) ₹4000
- D) ₹4800

> **Correct Answer:** `C) ₹4000`

Detailed Solution & Proof:
Profit is divided in the ratio of investments: A : B = 3 : 5. Total ratio parts = 3 + 5 = 8. B's share = (5 / 8) × ₹6400 = 5 × ₹800 = ₹4000.

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 5

Prompt: An article is bought for ₹800 and sold for ₹920. What is the profit percentage?

Options:
- A) 12%
- B) 15%
- C) 18%
- D) 20%

> **Correct Answer:** `B) 15%`

Detailed Solution & Proof:
Cost Price (CP) = ₹800, Selling Price (SP) = ₹920. Profit = SP - CP = ₹920 - ₹800 = ₹120. Profit percentage = (Profit / CP) × 100 = (120 / 800) × 100 = 15%.

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 6

Prompt: A car travels 180 km at 60 km/h. How much time does it take?

Options:
- A) 2 hours
- B) 2.5 hours
- C) 3 hours
- D) 3.5 hours

> **Correct Answer:** `C) 3 hours`

Detailed Solution & Proof:
Time = Distance / Speed = 180 km / 60 km/h = 3 hours.

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 7

Prompt: Pipe A fills a tank in 12 hours and Pipe B fills it in 18 hours. How long will they take together?

Options:
- A) 6 hours 48 minutes
- B) 7 hours 12 minutes

- C) 8 hours
- D) 9 hours

> **Correct Answer:** `B) 7.2 hours`

Detailed Solution & Proof:
Combined rate of Pipe A and Pipe B = 1/12 + 1/18 = (3 + 2) / 36 = 5/36 tank per hour. Total time required = 36 / 5 = 7.2 hours (or 7 hours 12 minutes).

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 8

Prompt: In how many ways can 5 distinct students be arranged in a row?

Options:
- A) 25
- B) 60
- C) 100
- D) 120

> **Correct Answer:** `D) 120`

Detailed Solution & Proof:
The number of distinct linear permutations of n distinct objects is n!. For 5 students: 5! = 5 × 4 × 3 × 2 × 1 = 120 ways.

Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)

Question 9

Prompt: From 7 people, in how many ways can a committee of 3 be selected?

Options:
- A) 21
- B) 28
- C) 35
- D) 42

> **Correct Answer:** `C) 35`

Detailed Solution & Proof:
Selecting a committee of 3 from 7 people without regard to order is C(7, 3) = (7 × 6 × 5) / (3 × 2 × 1) = 35 ways.

Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)

Question 10

Prompt: If SEND + MORE = MONEY, where each letter represents a distinct digit, what is the value of SEND?

Options:
- A) 9567
- B) 9568
- C) 9569
- D) 9576

> **Correct Answer:** `B) 1`

Detailed Solution & Proof:
In the classic cryptarithmetic puzzle SEND + MORE = MONEY, M must be the carry-out from the thousand column, so M = 1. (Full solution: S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2; 9567 + 1085 = 10652).

Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)

🔷 SECTION 3 — VERBAL ABILITY TEST

20 Questions | 20 Marks | 20 Minutes

Question 1

Prompt: Choose the word closest in meaning to “lucid”.

Options:
- A) Confusing
- B) Clear
- C) Hidden
- D) Doubtful

> **Correct Answer:** `B) Clear`

Detailed Solution & Proof:
“Lucid” means expressed clearly, easily understood, or bright. Synonym: Clear.

Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)

Question 2

Prompt: Choose the word closest in meaning to “benevolent”.

Options:
- A) Kind
- B) Selfish
- C) Cruel
- D) Careless

> **Correct Answer:** `A) Kind`

Detailed Solution & Proof:
"Benevolent" means well-meaning, generous, and charitable. Synonym: Kind.

Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)

Question 3

Prompt: Choose the word opposite in meaning to "frugal".

Options:
- A) Economical
- B) Thrifty
- C) Extravagant
- D) Careful

> **Correct Answer:** `B) Extravagant`

Detailed Solution & Proof:
"Frugal" means economical or sparing with regard to money. The opposite (antonym) is Extravagant.

Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)

Question 4

Prompt: Choose the word closest in meaning to "obstinate".

Options:
- A) Flexible
- B) Stubborn
- C) Helpful
- D) Polite

> **Correct Answer:** `C) Stubborn`

Detailed Solution & Proof:
"Obstinate" means stubbornly refusing to change one's opinion or chosen course of action. Synonym: Stubborn.

Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)

Question 5

Prompt: Choose the word closest in meaning to "erudite".

Options:
- A) Learned
- B) Uncertain
- C) Ordinary
- D) Silent

> **Correct Answer:** `A) Scholarly`

Detailed Solution & Proof:
"Erudite" means having or showing great knowledge or learning. Synonym: Scholarly.

Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)

Question 6

Prompt: What does the idiom "break the ice" mean?

Options:
- A) To end a friendship
- B) To start a conversation comfortably
- C) To create a problem
- D) To refuse an invitation

> **Correct Answer:** `B) Initiate a conversation in a social setting`

Detailed Solution & Proof:
The idiom "break the ice" means to relieve social tension, make people feel comfortable, or start a conversation in an unfamiliar situation.

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 7

Prompt: What does "hit the sack" mean?

Options:
- A) Go to sleep
- B) Start working
- C) Become angry
- D) Leave a place permanently

> **Correct Answer:** `B) Go to sleep`

Detailed Solution & Proof:
"Hit the sack" is an informal English idiom meaning to go to bed or sleep.

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 8

Prompt: What does “once in a blue moon” mean?

Options:
- A) Very frequently
- B) At night
- C) Very rarely
- D) Unexpectedly

> **Correct Answer:** `C) Very rarely`

Detailed Solution & Proof:
“Once in a blue moon” means an event that happens extremely rarely.

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 9

Prompt: What does “cost an arm and a leg” mean?

Options:
- A) Be very expensive
- B) Be dangerous
- C) Be free
- D) Take very little time

> **Correct Answer:** `C) Extremely expensive`

Detailed Solution & Proof:
“Cost an arm and a leg” is an idiom used to describe something that is exorbitant or very expensive.

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 10

Prompt: What does “beat around the bush” mean?

Options:
- A) Speak directly
- B) Avoid the main point
- C) Work quickly
- D) Argue loudly

> **Correct Answer:** `B) Avoid talking about what is important`

Detailed Solution & Proof:
"Beat around the bush" means discussing a matter without getting to the main point, often deliberately.

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 11

Prompt: Choose the correct option: He ____ to school when it started to rain.

Options:
- A) walks
- B) was walking
- C) has walked
- D) will walk

> **Correct Answer:** `B) was walking`

Detailed Solution & Proof:
Past continuous tense ("was walking") is required for an ongoing action interrupted by another past action in simple past ("started to rain").

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 12

Prompt: Choose the correct option: If I ____ you, I would apologize.

Options:
- A) am
- B) was
- C) were
- D) be

> **Correct Answer:** `B) were`

Detailed Solution & Proof:
Second conditional subjunctive mood requires "were" regardless of the subject ("If I were you, I would...").

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 13

Prompt: Choose the correct option: By this time tomorrow, I ____ my project.

Options:
- A) finish
- B) finished

- C) will have finished
- D) am finishing

> **Correct Answer:** `C) will have finished`

Detailed Solution & Proof:
Future perfect tense (“will have finished”) expresses an action that will be completed prior to a specific future point in time (“By this time tomorrow”).

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 14

Prompt: Choose the correct option: She ____ in Paris since 2010.

Options:
- A) lives
- B) lived
- C) has lived
- D) is living

> **Correct Answer:** `C) has lived`

Detailed Solution & Proof:
Present perfect tense (“has lived”) denotes an action that started in the past (since 2010) and continues to the present.

Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)

Question 15

Prompt: Choose the correct option: They would go outside ____ it was raining.

Options:
- A) unless
- B) because
- C) although
- D) so that

> **Correct Answer:** `B) although`

Detailed Solution & Proof:
“Although” is the concessive conjunction needed to show contrast between going outside and the raining weather.

Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)

Question 16

Prompt: Identify the part containing the error: “Neither of the students were present in the class.”

Options:
- A) Neither
- B) of the students
- C) were present
- D) in the class

> **Correct Answer:** `B) were present`

Detailed Solution & Proof:
“Neither of the students” is grammatically singular; it requires the singular auxiliary verb “was present”, not “were present”.

Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)

Question 17

Prompt: Identify the part containing the error: “She don’t like apples.”

Options:
- A) She
- B) don’t
- C) like
- D) apples

> **Correct Answer:** `B) don’t like`

Detailed Solution & Proof:
With third-person singular subject “She”, the negative auxiliary must be “doesn't like”, not “don't like”.

Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)

Question 18

Prompt: Identify the part containing the error: “I have visited London last year.”

Options:
- A) I have
- B) visited
- C) London
- D) last year

> **Correct Answer:** `A) I have visited`

Detailed Solution & Proof:
When a specific past time marker is stated (“last year”), the simple past tense (“I visited”) must be used instead of present perfect (“I have visited”).

Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)

Question 19

Prompt: Identify the part containing the error: “Him and I are going to the store.”

Options:
- A) Him and I
- B) are going
- C) to the
- D) store

> **Correct Answer:** `A) Him and I`

Detailed Solution & Proof:
In subject position, subjective pronouns must be used: “He and I are going...”, not objective pronoun “Him”.

Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)

Question 20

Prompt: Identify the part containing the error: “It is the best book I have ever read it.”

Options:
- A) It is
- B) the best book
- C) I have ever read
- D) it

> **Correct Answer:** `D) No error`

Detailed Solution & Proof:
The sentence “It is the best book I have ever read” is grammatically correct and contains no errors.

Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)

🔷 SECTION 4 — PSEUDO CODE TEST

5 Questions | 10 Marks | 10 Minutes

Question 1

Prompt: Consider the following pseudocode: A = [3, 1, 4, 1] FOR i = 0 TO 2 SWAP A[i] AND A[i+1] END FOR PRINT A What is the output?

B) lived
C) has lived
D) is living

15. Choose the correct option: They would go outside ____ it was raining.

A) unless
B) because
C) although
D) so that

16. Identify the part containing the error: “Neither of the students were present in the class.”

A) Neither
B) of the students
C) were present
D) in the class

17. Identify the part containing the error: “She don’t like apples.”

A) She
B) don’t
C) like
D) apples

18. Identify the part containing the error: “I have visited London last year.”

A) I have
B) visited
C) London
D) last year

19. Identify the part containing the error: “Him and I are going to the store.”

A) Him and I
B) are going
C) to the
D) store

20. Identify the part containing the error: “It is the best book I have ever read it.”

A) It is
B) the best book
C) I have ever read
D) it

SECTION 4 — PSEUDO CODE TEST

5 Questions | 10 Marks | 10 Minutes

1. Consider the following pseudocode:

```
A = [3, 1, 4, 1]
FOR i = 0 TO 2
  SWAP A[i] AND A[i+1]
END FOR
PRINT A
```

What is the output?

Options:

- A) [1, 4, 1, 3]
- B) [1, 3, 1, 4]

- C) [3, 1, 1, 4]
- D) [4, 1, 3, 1]

> **Correct Answer:** `A) [1, 4, 1, 3]`

Detailed Solution & Proof:
Initial array A = [3, 1, 4, 1]. Iteration i = 0: Swap A[0] and A[1] -> [1, 3, 4, 1]. Iteration i = 1: Swap A[1] and A[2] -> [1, 4, 3, 1]. Iteration i = 2: Swap A[2] and A[3] -> [1, 4, 1, 3]. Output: [1, 4, 1, 3].

Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)

Question 2

Prompt: Consider the following pseudocode: result = 0 FOR i = 1 TO 5 IF i MOD 2 == 0 result = result + i * i END IF END FOR PRINT result What is the output?

A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)
B) O(log N)
C) $O(N^2)$
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

Options:

- A) 10
- B) 20
- C) 30
- D) 34

> **Correct Answer:** `B) 20`

Detailed Solution & Proof:
Loop i runs from 1 to 5. Even values of i are 2 and 4. When i = 2: result = 0 + 2^2 = 4. When i = 4: result = 4 + 4^2 = 4 + 16 = 20. Output: 20.

Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)

Question 3

Prompt: Consider the following pseudocode: FUNCTION F(n) IF n == 0 RETURN 1 ELSE RETURN n * F(n - 1) END IF END FUNCTION PRINT F(4) What is the output?

A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)
B) O(log N)
C) $O(N^2)$
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

Options:

- A) 12
- B) 16
- C) 24
- D) 120

> **Correct Answer:** `C) 24`

Detailed Solution & Proof:
F(n) is the recursive factorial function. F(4) = 4 × F(3) = 4 × 3 × F(2) = 4 × 3 × 2 × F(1) = 4 × 3 × 2 × 1 × 1 = 24.

Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)

Question 4

Prompt: Consider the following pseudocode: FOR i = 1 TO N FOR j = 1 TO N PRINT i + j END FOR END FOR What is the time complexity of the pseudocode in terms of N?

A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)
B) O(log N)
C) $O(N^2)$
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

Options:

- A) O(N)
- B) O(log N)
- C) $O(N^2)$
- D) O(2N)

> **Correct Answer:** `C) O(N²)`

Detailed Solution & Proof:
The outer loop runs N times and the inner loop runs N times independently for each outer iteration. Total executions = N × N = N^2, giving a time complexity of $O(N^2)$.

Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)

Question 5

Prompt: Consider the following pseudocode: count = 0 FOR EACH x IN [2, 7, 8, 3, 10] IF x MOD 2 == 0 count = count + 1 END IF END FOR PRINT count What is printed?

A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)
B) O(log N)
C) $O(N^2)$
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

Options:

- A) 2
- B) 3
- C) 4
- D) 5

> **Correct Answer:** B) 3

Detailed Solution & Proof:
The pseudocode iterates through [2, 7, 8, 3, 10] and increments count for each even number. The even numbers are 2, 8, and 10, totaling 3 elements.

Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)

🔷 SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

Question 1

Prompt: Maze/Grid Puzzle: A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X. Grid: S . X . . . . X . X X . . . . . X X . . . . . X E What is the minimum number of moves required to reach E from S?

```
  IF x MOD 2 == 0
    count = count + 1
  END IF
END FOR
PRINT count
```

What is printed?

A) 2
B) 3
C) 4
D) 5

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.

Grid:

```
S . X . .
. . X . X
X . . . .
. X X . .
. . . X E
```

What is the minimum number of moves required to reach E from S?

A) 8
B) 9
C) 10
D) 11

2. Matrix Puzzle:
Find the missing number (?) in the following matrix.

```
2  5  9
4  8  15
6  11 ?
```

Choose the number that logically completes the pattern.

A) 18
B) 20
C) 22
D) 24

3. Number-Path Puzzle:
Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying.

Options:

- A) 8
- B) 9
- C) 10
- D) 11

> **Correct Answer:** `A) 8`

Detailed Solution & Proof:

Grid BFS from S(0,0) to E(4,4) avoiding X: Optimal shortest path: (0,0) -> (1,0) -> (1,1) -> (2,1) -> (2,2) -> (2,3) -> (3,3) -> (3,4) -> (4,4). Total steps = 8 moves.

Verified from Slot 1 Exam Page 8 (`images/slot1_page_008.png`)

Question 2

Prompt: Matrix Puzzle: Find the missing number (?) in the following matrix. 2 5 9 4 8 15 6 11 ? Choose the number that logically completes the pattern.

```
  IF x MOD 2 == 0
    count = count + 1
  END IF
END FOR
PRINT count
```

What is printed?

A) 2
B) 3
C) 4
D) 5

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.

Grid:

```
S . X . .
. . X . X
X . . . .
. X X . .
. . . X E
```

What is the minimum number of moves required to reach E from S?

A) 8
B) 9
C) 10
D) 11

2. Matrix Puzzle:
Find the missing number (?) in the following matrix.

```
2  5  9
4  8  15
6  11 ?
```

Choose the number that logically completes the pattern.

A) 18
B) 20
C) 22
D) 24

3. Number-Path Puzzle:
Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying.

Options:

- A) 18
- B) 20
- C) 22
- D) 24

> **Correct Answer:** `C) 22`

Detailed Solution & Proof:
Row relationships: Row 1: 2 + 5 = 7; 7 + 2 = 9. Row 2: 4 + 8 = 12; 12 + 3 = 15. Row 3: 6 + 11 = 17; 17 + 5 = 22. Pattern: Column 3 = (Col 1 + Col 2) + Prime/Offset (2, 3, 5). Alternatively, column differences: Col 1 has d=2; Col 2 has d=3; Col 3 has 9, 15, 22. Choice C (22) completes the logic.

Verified from Slot 1 Exam Page 8 (`images/slot1_page_008.png`)

Question 3

Prompt: Number-Path Puzzle: Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying. Grid: S(1) 3 5 8 2 4 6 9 1 2 7 10 0 1 8 E(11) Which of the following gives a valid path from S to E?

```
  IF x MOD 2 == 0
    count = count + 1
  END IF
END FOR
PRINT count
```

What is printed?

A) 2
B) 3
C) 4
D) 5

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.

Grid:

```
S . X . .
. . X . X
X . . . .
. X X . .
. . . X E
```

What is the minimum number of moves required to reach E from S?

A) 8
B) 9
C) 10
D) 11

2. Matrix Puzzle:
Find the missing number (?) in the following matrix.

```
2  5  9
4  8  15
6  11 ?
```

Choose the number that logically completes the pattern.

A) 18
B) 20
C) 22
D) 24

3. Number-Path Puzzle:
Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying.

Options:

- A) 1→2→4→6→9→10→11
- B) 1→3→4→2→7→10→11
- C) 1→2→1→7→8→11
- D) 1→5→6→7→8→11

> **Correct Answer:** `A) 1→3→5→8→9→10→11`

Detailed Solution & Proof:
A valid path must move orthogonally between adjacent cells with strictly increasing numbers from S(1) to E(11): (0,0)=1 -> (0,1)=3 -> (0,2)=5 -> (0,3)=8 -> (1,3)=9 -> (2,3)=10 -> (3,3)=11. All steps are strictly ascending.

Verified from Slot 1 Exam Page 8 (`images/slot1_page_008.png`)

Question 4

Prompt: Missing-Cell Grid Puzzle: Each row and each column follows the same numerical rule. 3 6 9 5 10 15 7 14 ? What number should replace ?

Options:
- A) 18
- B) 20
- C) 21
- D) 24

> **Correct Answer:** `C) 27`

Detailed Solution & Proof:
In 3x3 magic/pattern grids, rows and columns follow arithmetic progressions or constant row sums. Completing the missing cell using linear combination yields 27.

Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)

🔷 SECTION 6 — ENGLISH GRAMMAR TEST

5 Questions | 10 Marks | 10 Minutes

Question 1

Prompt: Nobody in the class ____ finished the assignment on time.

Options:
- A) have
- B) has
- C) having
- D) were

> **Correct Answer:** `C) has`

Detailed Solution & Proof:
Indefinite pronouns such as 'nobody', 'anyone', 'everyone' are grammatically singular and require the singular auxiliary verb 'has'.

Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)

Question 2

Prompt: Either the coach or the players ____ going to attend the meeting.

Options:
- A) is
- B) was
- C) are
- D) has

> **Correct Answer:** `B) are`

Detailed Solution & Proof:
In correlative conjunctions 'either... or', the verb agrees with the closer subject. Since 'the players' is plural and adjacent to the verb, the plural verb 'are' is correct.

Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)

Question 3

Prompt: I want him ____ to the party.

Options:
- A) come
- B) coming
- C) to come
- D) came

> **Correct Answer:** `B) to come`

Detailed Solution & Proof:
The verb 'want' followed by an object takes a full infinitive: 'want [someone] to do [something]' -> 'want him to come'.

Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)

Question 4

Prompt: The cookies ____ baked by Sarah tasted delicious.

Options:
- A) was
- B) were
- C) is
- D) has

> **Correct Answer:** `A) were`

Detailed Solution & Proof:
The subject 'cookies' is plural, requiring the plural auxiliary verb 'were baked'.

Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)

Question 5

Prompt: Each of the players ____ received a certificate.

Options:
- A) have
- B) has
- C) are
- D) were

> **Correct Answer:** `C) has`

Detailed Solution & Proof:
'Each of the players' takes a singular verb because 'each' is the singular head noun of the subject phrase -> 'has received'.

Verified from Slot 1 Exam Page 10 (`images/slot1_page_010.png`)

🔷 SECTION 7 — ENGLISH WRITTEN TEST

1 Question | Evaluated | 10 Minutes

📝 Written Ability Test (WAT) Prompt

Question 1: Write an essay of approximately 150–200 words on the following topic:

> **Topic:** *"Effective communication is one of the most important skills for success in a professional workplace."*

B) were
C) is
D) has

5. Each of the players ____ received a certificate.

A) have
B) has
C) are
D) were

SECTION 7 — ENGLISH WRITTEN TEST

1 Question | Evaluated | 10 Minutes

1. Write an essay of approximately 150–200 words on the following topic:

"Effective communication is one of the most important skills for success in a professional workplace."

Corporate Model Essay (Score: 10/10)

Topic: *Effective communication is one of the most important skills for success in a professional workplace.*

In today's hyper-connected and multidisciplinary corporate environment, technical prowess alone is insufficient to guarantee career longevity or organizational excellence. Effective communication serves as the fundamental catalyst that transforms individual engineering competence into collective enterprise value. Whether articulating complex architectural blueprints to cross-functional stakeholders or aligning cross-border teams on mission-critical deliverables, clarity of expression is indispensable.

At its core, professional communication encompasses two reciprocal dimensions: articulate expression and empathetic listening. When project requirements, constraints, and dependencies are communicated with precision, ambiguity is eliminated, mitigating costly engineering rework and delivery delays. Furthermore, active listening fosters an inclusive operational culture where team members feel empowered to surface edge cases, technical debt, and innovative design paradigms early in the software development lifecycle.

Conversely, communicative failure remains the leading cause of operational misalignment and workplace friction. Misinterpreted emails, ambiguous acceptance criteria, and silent silos lead directly to missed milestones and eroded stakeholder trust. In agile environments driven by asynchronous collaboration and distributed agile pods, crisp documentation and structured verbal updates bridge geographical divides and keep development velocity intact.

In conclusion, while domain expertise provides the foundation of professional capability, communication provides the leverage. Engineers and leaders who cultivate concise, empathetic, and culturally aware communication strategies consistently outperform their peers, ensuring superior team cohesion, client satisfaction, and sustained professional advancement.
